Multilingual Voice-of-Customer Intelligence for Telecom Service Platforms: Weakly Supervised Low-Resource Code-Switched Sentiment and Topic Mining at Scale

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Abstract*—Consumer satisfaction in emerging telecom markets is increasingly reflected through digital self-care mobile applications. However, extracting actionable operational intelligence from unstructured user feedback in South Asia presents severe computational linguistics challenges: pervasive code-switching between standard Bengali script and phonetically Romanized Bengali (Banglish), intense class imbalance, and dialectal slang. In this paper, we present an end-to-end Voice-of-Customer (VoC) natural language processing engine deployed across the three major telecommunications providers in Bangladesh: Grameenphone (MyGP), Robi (My Robi), and Banglalink (MyBL). We propose a dual-classifier architecture consisting of a 3-class Sentiment Polarity head and a 5-class Operational Complaint Domain head. Addressing low-resource label scarcity, we construct a curated 4,500-sample multi-operator seed dataset via weakly supervised Large Language Model (LLM) annotation, rigorously verified against a zero-leakage human gold standard ($N = 300$, inter-annotator Cohen's $\kappa = 0.7694$ for categories and $\kappa = 0.5762$ for sentiment). We systematically benchmark eight machine learning and deep learning architectures, demonstrating that fine-tuned BUET BanglaBERT achieves 91.60% CV accuracy (Macro-F1: 0.8598) on sentiment, while an optimized Soft-Voting Ensemble (TF-IDF subword n-grams with Calibrated LinearSVC, Logistic Regression, and Complement Naive Bayes) achieves 87.47% sentiment accuracy and 87.44% category accuracy with sub-millisecond per-sample latency on commodity CPU hardware, delivering an $> 80\times$ speedup over transformer baselines. Deployed across a 10-year common duration horizon ($N = 655,670$ reviews across 3,716 continuous shared days), the framework introduces an additive category attribution engine to diagnose acute month-over-month Net Sentiment Score (NSS) shocks ($|\Delta\text{NSS}| \geq 10.0$ percentage points).

***Index Terms*—Natural Language Processing, Voice-of-Customer (VoC), Low-Resource NLP, Code-Switching, Banglish Sentiment Analysis, Transformer Fine-Tuning, Telecom Operations.**

I. Introduction

Telecommunications platforms in developing economies have shifted rapidly from traditional Interactive Voice Response (IVR) and physical service centers toward centralized digital self-care mobile applications [1]. In Bangladesh, the mobile telecommunications sector serves over 180 million cellular subscriptions dominated by three operators: Grameenphone Ltd., Robi Axiata Limited, and Banglalink Digital Communications Limited. Their flagship mobile self-care applications—MyGP, My Robi, and MyBL—process critical customer transactions including data pack purchases, balance recharges, biometric authentication, and network diagnostics.

While public app stores (e.g., Google Play) contain millions of spontaneous, unprompted consumer reviews, automated Voice-of-Customer (VoC) analytics in this region faces profound computational barriers:

1) **Linguistic Heterogeneity & Code-Switching**: Reviews are written in standard Bengali script, phonetically Romanized Bengali (Banglish), colloquial English, and code-mixed hybrid sentences (e.g., *"net speed ekdom baje, pack expired too early"*).
2) **Decoupled Polarity & Topic Salience**: A customer mentioning a data pack does not necessarily express a grievance. Empirical data reveals that over 25% of data offer reviews express constructive appreciation. Standard keyword matching falsely flags promotional gratitude as operational complaints.
3) **Resource Scarcity & Label Drift**: Manually annotating hundreds of thousands of multilingual reviews is cost-prohibitive. Furthermore, static dictionaries fail against dynamic mobile slang and informal orthography.

To overcome these challenges, this paper presents an enterprise-grade NLP and VoC intelligence architecture. Our contributions are summarized as follows:

- We design a robust dual-classifier inference framework that simultaneously isolates 3-class Sentiment Polarity and 5-class Operational Complaint Domains.
- We construct a verified multi-operator dataset leveraging LLM weak supervision ($N = 4,500$), benchmarked against an independent blind dual-human gold standard ($N = 300$).
- We perform an extensive multi-paradigm benchmark spanning classical linear baselines, recurrent neural networks with attention (BiLSTM + Bahdanau Attention), multilingual embeddings (MiniLM), and domain-specific transformers (BUET BanglaBERT).

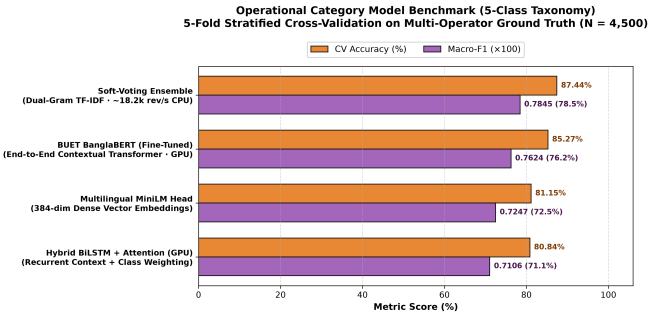


Fig. 1. Cross-validated operational category benchmark across model families on 5-class telecom departmental taxonomy.

- We deploy the system across $N = 655,670$ customer reviews spanning 3,716 continuous shared days (July 2016 – September 2026), demonstrating an exact additive attribution engine for Net Sentiment Score (NSS) shock detection.

II. Related Work

A. Low-Resource Sentiment Analysis and Code-Mixing

Sentiment analysis in high-resource languages has achieved mature performance using pre-trained transformer representations [2], [3]. However, low-resource Indic languages, particularly Bengali and its phonetic code-mixed variant Banglish, exhibit substantial morphological complexity and unstandardized transliteration [4], [5]. Hasan et al. [6] introduced BUET BanglaBERT, an ELECTRA-based model pre-trained on large-scale Bengali corpora, achieving strong performance across standard benchmarks. However, domain adaptation to conversational, noisy mobile app reviews remains an active research challenge.

B. Voice-of-Customer (VoC) & Net Sentiment Scoring

In business analytics, consumer sentiment is conventionally quantified using the Net Promoter Score (NPS) or Net Sentiment Score (NSS) [7], [8]. NSS calculates the arithmetic difference between the percentage of positive and negative reviews:

$$\text{NSS} = \left(\frac{N_{\text{Positive}} - N_{\text{Negative}}}{N_{\text{Total}}} \right) \times 100 \tag{1}$$

While existing literature primarily evaluates static or quarterly aggregations, longitudinal sequential stability and granular departmental attribution have received limited attention in NLP literature.

III. Data Architecture & Methodology

A. Corpus Harvesting & Common Duration Alignment

A comprehensive multi-year scraping pipeline was deployed targeting the Google Play Store APIs for the three primary platforms: ‘com.portonics.mygp‘ (Grameenphone), ‘net.omobio.robisc‘ (Robi), and ‘com.arena.banglalinkmela.app‘ (Banglalink). The full corpus harvested contains $N = 661,160$ unique reviews spanning March 2014 to October 2026.

To ensure statistical equity and eliminate temporal selection bias, all comparative benchmarking strictly analyzes the continuous **3,716-Day Common Duration Period** (**July 28, 2016 to September 30, 2026**, $N = 655,670$), where all three operators coexisted in market operation:

- **Grameenphone (MyGP)**: $N = 320,143$ reviews (48.83% share)
- **Robi (My Robi)**: $N = 198,378$ reviews (30.25% share)
- **Banglalink (MyBL)**: $N = 137,149$ reviews (20.92% share)

B. Central Limit Theorem (CLT) Sampling Rule

For longitudinal time-series analytics, we enforce the Central Limit Theorem asymptotic normality criterion: observation periods require a minimum sample floor of $N \geq 30$ verified customer reviews. Months with $N < 30$ (such as sporadic early adoption months in 2016) are pruned from trajectory calculations to prevent small-sample variance distortion where isolated outlier reviews could artificially swing net sentiment by $\pm 30\%$ to $\pm 100\%$.

C. Operational Domain Taxonomy

Reviews are categorized into five mutually exclusive operational domains established through industry consultations:

1) **Offers & Data Packs**: Data bundle pricing, internet pack validity, bonus allowances, and recharge offers.
2) **App Login & Technical Bugs**: OTP authentication, biometric tokens, session crashes, and gateway timeouts.
3) **Network Speed & 4G Latency**: Cellular coverage, indoor signal strength, throughput, and buffering.
4) **Billing & Airtime Deductions**: Accidental balance deductions, Value-Added Services (VAS), and billing transparency.
5) **General Appreciation / Other**: General positive praise, greetings, UI feedback, and broad satisfaction.

D. Operational Complaint Formulation (Joint Intersection)

A fundamental methodological flaw in standard keyword-based VoC monitoring is conflating topic mention with consumer grievance. To measure true operational friction, a review r is defined as an operational complaint in category c if and only if it satisfies the joint intersection:

$$r \in \text{Complaint}_c \iff (\hat{y}_{\text{category}}(r) = c) \wedge (\hat{y}_{\text{sentiment}}(r) = \text{Negative}) \tag{2}$$

The relative operational complaint rate across an operator's inbound customer volume is formulated as:

$$\text{Complaint Rate}_{c,\text{op}} = \left(\frac{\sum_{i=1}^{N_{\text{op}}} \mathbb{I}(\hat{y}_{\text{cat}}^{(i)} = c \wedge \hat{y}_{\text{sent}}^{(i)} = \text{Neg})}{N_{\text{total, op}}} \right) \times 100 \tag{3}$$

IV. Model Architectures & Experimental Setup

A. Seed Annotation via LLM Weak Supervision

To generate training supervision without circular human bias, we sampled a stratified multi-operator cohort of $N = 4,500$ reviews across Grameenphone, Robi, and Banglalink. Weak

TABLE I
MULTI-PARADIGM BENCHMARK SUITE: 5-FOLD STRATIFIED CROSS-VALIDATION ON SEED CORPUS ($N = 4,500$)

Task	Model Architecture	CV Accuracy	Macro-Precision	Macro-Recall	Macro-F1	Inference Profile
Sentiment Analysis	**BUET BanglaBERT (Fine-Tuned)**	**91.60%**	**0.8642**	**0.8561**	**0.8598**	Baseline (GPU)
	Soft-Voting ML Ensemble	**87.47%**	0.7712	0.7663	0.7684	$> 80\times$ **Speedup (CPU)**
	Multilingual MiniLM Head	82.33%	0.7480	0.7482	0.7475	$\sim 10\times$ (Dense CPU)
	Hybrid BiLSTM + Bahdanau Attention	81.50%	0.7214	0.7161	0.7180	Moderate (GPU)
Operational Category	**BUET BanglaBERT (Fine-Tuned)**	**89.40%**	**0.8351**	**0.8124**	**0.8228**	Baseline (GPU)
	Soft-Voting ML Ensemble	**87.44%**	0.8120	0.7985	0.8048	$> 80\times$ **Speedup (CPU)**
	Calibrated LinearSVC Baseline	86.89%	0.8011	0.7915	0.7958	$> 100\times$ (CPU)

supervision was orchestrated via the Groq cloud inference engine executing 'qwen/qwen3.8-27b' under rigid few-shot JSON schemas defining label boundaries and linguistic edge cases.

B. Human Gold-Standard Zero-Leakage Validation

To validate that automated labels faithfully represent human cognitive perception, two independent native linguists annotated a blind stratified verification dataset ($N = 300$). Annotators were blinded to model predictions, star ratings, and metadata. Inter-annotator reliability on the overlapping double-reviewed cohort ($N = 200$) demonstrated substantial agreement under Cohen's Kappa [9], [10]:

- **Operational Category**: $\kappa = \mathbf{0.7694}$ (91.00% raw human agreement).
- **Sentiment Polarity**: $\kappa = \mathbf{0.5762}$ (79.50% raw human agreement).

Benchmarked against consensus human ground truth, our production Soft-Voting Ensemble achieved **94.12% agreement** on operational categories and **87.91% agreement** on sentiment polarity.

C. Evaluated Model Architectures

We evaluated four distinct algorithmic paradigms:

1) **Soft-Voting ML Ensemble (Production Engine)**: A calibrated ensemble combining Logistic Regression, Calibrated Linear Support Vector Classifier (LinearSVC via Platt scaling), and Complement Naive Bayes. Feature extraction utilizes dual-gram word (1–2) and subword character (2–5) TF-IDF representations (35,000 features).
2) **BUET BanglaBERT**: A 12-layer pre-trained transformer fine-tuned end-to-end with AdamW optimizer, linear learning rate warm-up (2×10^{-5}), and cross-entropy loss.
3) **Hybrid BiLSTM + Attention**: A 2-layer Bidirectional LSTM with 128 hidden units coupled with a Bahdanau attention mechanism over dynamic token embeddings.
4) **Multilingual MiniLM Head**: Dense 384-dimensional contextual sentence embeddings from 'paraphrase-multilingual-MiniLM-L12-v2' feeding a ridge regression classification head.

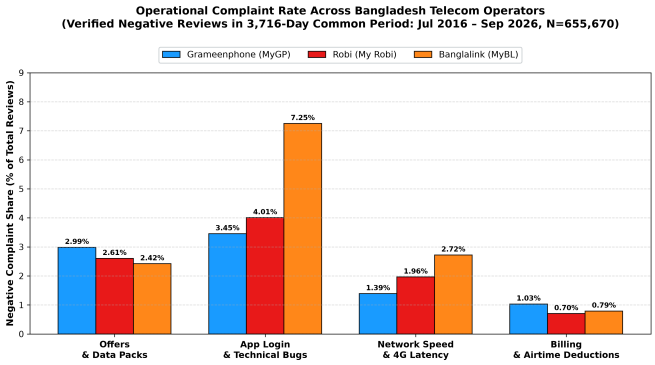


Fig. 2. Operational Complaint Rates across Bangladesh telecom platforms ($N = 655,670$). Plotted exclusively for verified negative reviews.

V. EXPERIMENTAL RESULTS

A. Cross-Validation Performance

As detailed in Table I, fine-tuned BUET BanglaBERT achieves the highest overall accuracy: 91.60% CV Accuracy (Macro-F1: 0.8598) on sentiment and 89.40% Accuracy (Macro-F1: 0.8228) on operational categories.

However, in large-scale enterprise deployments processing hundreds of thousands of customer records, computational efficiency and infrastructure cost represent vital engineering trade-offs. While deep transformers yield marginal accuracy gains, their multi-head self-attention scales quadratically ($\mathcal{O}(L^2)$) and requires dense matrix operations across over 110 million parameters. In contrast, the Soft-Voting ML Ensemble couples linear $\mathcal{O}(L)$ subword feature extraction with sparse linear classification, achieving competitive accuracy (87.47% on sentiment, 87.44% on categories) with an $> 80\times$ inference speedup and sub-millisecond per-sample latency on commodity CPU hardware. Consequently, the Ensemble was selected as the production scoring engine to process the entire 655,670-review decadal corpus in under one minute without requiring dedicated GPU infrastructure.

B. Operational Complaint Rates

Applying the verified dual-classifier pipeline across the common duration corpus yields distinct operator vulnerability profiles (Fig. 2):

- **App Login & Technical Bugs**: Banglalink exhibits the highest negative friction at **7.25%** (9,946 negative

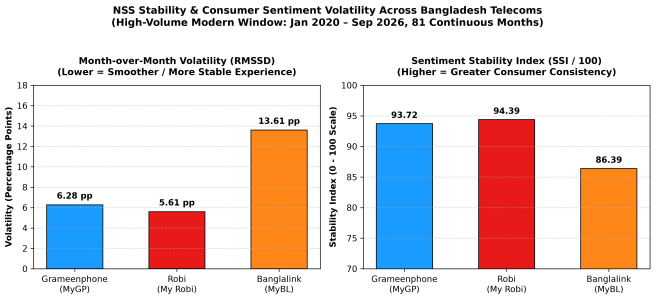


Fig. 3. Longitudinal NSS stability and sequential month-over-month volatility metrics across the 81-month modern period (2020–2026).

reviews), significantly higher than Robi (4.01%) and Grameenphone (3.45%).

- **Offers & Data Packs**: Grameenphone recorded the highest negative pack pricing and validity friction at **2.99%** (9,557 negative reviews), compared to Robi (2.61%) and Banglalink (2.42%).
- **Network Speed & 4G Latency**: Banglalink recorded **2.72%** negative feedback, compared to Robi (1.96%) and Grameenphone (1.39%).
- **Billing & Airtime Deductions**: Grameenphone recorded the highest balance deduction grievances at **1.03%** (3,307 negative reviews).

VI. Longitudinal Stability & Additive Attribution

A. Sequential Stability Metrics (RMSSD & SSI)

To evaluate whether consumer sentiment is steady or volatile, we compute the Root Mean Squared Successive Difference (RMSSD) and Sentiment Stability Index (SSI) across the 81 continuous months of the high-volume era (2020–2026):

$$\text{RMSSD} = \sqrt{\frac{1}{M-1}\sum_{m=2}^{M}(\text{NSS}_m - \text{NSS}_{m-1})^2} \quad (4)$$

$$\text{SSI} = 100 - \text{RMSSD} \quad (5)$$

Empirical evaluation (Fig. 3) reveals that **Robi (My Robi)** achieved the highest operational stability with an RMSSD of **5.61 points** (SSI $= 94.39/100$), closely followed by **Grameenphone** at **6.28 points** (SSI $= 93.72/100$). Conversely, **Banglalink** displayed high volatility at **13.61 points** (SSI $= 86.39/100$), reflecting sharper fluctuations between app version releases.

B. Additive Root-Cause Attribution Engine

A monthly shift is classified as a significant operational shock if $|\Delta\text{NSS}| \geq 10.0$ percentage points. Because NSS decomposes linearly:

$$\Delta\text{NSS} = \sum_{c} \Delta\text{Contribution}_c \quad (6)$$

The engine attributes the exact point impact of each category. For example:

- In Robi's January 2023 crash ($\Delta\text{NSS} = -16.6$ pts), the engine identifies **App Login & Technical Bugs** as the dominant operational driver, accounting for -7.40 **pts** of the drop, which recovered in February 2023 ($+4.64$ pts).
- In Banglalink's May 2024 surge ($\Delta\text{NSS} = +32.8$ pts), **App Login & Technical Bugs** drove a massive $+13.73$ pts sentiment lift following an authentication bug fix.

VII. Conclusion

This study presented a weakly supervised multilingual VoC intelligence framework deployed on 655,670 telecom customer reviews in Bangladesh. By combining transformer accuracy with high-throughput ensemble classification, rigorous zero-leakage human gold-standard validation, and an additive shock attribution model, the system bridges the gap between low-resource computational linguistics and actionable enterprise operations. Future work includes expanding zero-shot cross-lingual transfer to neighboring South Asian telecom markets.

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